a) Initial Student ID: 12521467

Array: [7, 6, 5, 4, 2, 2, 1, 1]

List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]

Insert 4 into an array:

Initial array: [7, 6, 5, 4, 2, 2, 1, 1]

Step 0:

Array: [7, 6, 5, 4, 2, 2, 1, 1]

Current element: ^

Current index: 0

Comment: if 7 > 4 ⇒ True, increment index

Step 1

Array: [7, 6, 5, 4, 2, 2, 1, 1]

Current element: ^

Current index: 1

Comment: if 6 > 4 ⇒ True, increment index

Step 2

Array: [7, 6, 5, 4, 2, 2, 1, 1]

Current element: ^

Current index: 2

Comment: if 5 > 4 ⇒ True, increment index.

Step 3

Array: [7, 6, 5, 4, 2, 2, 1, 1]

Current element: ^

Current index: 3

Comment: if 4 > 4 ⇒ False Insertion point found at index 3.

Final array: [7, 6, 5, 4, 4, 2, 2, 1, 1],
^
inserted value

when we find the insertetion point we make space for the new element, so we place every next element starting from index 3 to the right.

And write the new number into the free slot at index 3.

6) Initial list [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]
Goal is to insert 4.

Insert 4 into a list.
Initial list: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]

Step 0:
List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]
Current element: ^
Comment: if 4 > 1 → True, next element by following the pointer.

Step 1:
List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]
Current element: ^
Comment: if 4 > 1 → True, next element by following the pointer

Step 2:
List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]
Current element: ^
Comment: if 4 > 2 → True, next element by following the pointer

Step 3:

List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]

Current element: ^

Comment: if 4 > 2 → True, next element by following the pointer

Step 4

List: [1] → [1] → [2] → [2] → [4] → [5] → [6] → [7]

Current element: ^

Comment: if 4 > 4 → False, insertion point found.

To insert 4 in a linked list we create a new node with value 4. Delete the chain link between [2] and [4] and place new node there. ~~and~~ Set (→) pointer of [2] pointing to the new node [4], and set (→) pointer of ~~a~~ new node pointing to [4].

new pointer   new element   new pointer

Final list: [1] → [1] → [2] → [2] → [4] → [4] → [5] → [6] → [7]